

# Tide retrospective: the analytic bridge in $\mathbb{R}^d$

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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## 1 Setting

The germbij identifiability arc closed with two structurally parallel chains, but their entry points differed: the one-dimensional chain starts from analyticity (`AnalyticAt` plus finite vanishing order), while the multivariate chain started from Taylor-structure hypotheses (a homogeneous leading part  $P$  with a remainder bound  $|g(x) - P(x)| \leq C\|x\|^{m+1}$ ). This tide supplies the missing bridge: deriving those Taylor hypotheses from a power series, so the multivariate turnkey theorem<sup>1</sup> can be invoked from analytic data alone, and the two chains match end to end.

## 2 The thing formalised

Two theorems in `Laplace/Multi/AnalyticBridge.lean`, sorry-free with zero warnings. The bridge<sup>2</sup>: if  $g$  has a power series  $p$  on a ball about 0 in  $\mathbb{R}^d$  and the diagonal evaluations  $p_k(x, \dots, x)$  vanish for all  $k < m$ , then with  $P(x) = p_m(x, \dots, x)$  there are  $C \geq 0$  and  $u_1 > 0$  with

$$|g(x) - P(x)| \leq C \|x\|^{m+1} \quad \text{for } \|x\| \leq 2u_1.$$

The composed corollary<sup>3</sup>: for  $L_1, L_2 \geq 0$  continuous with  $L_2 - L_1$  admitting such a series with  $P \neq 0$  somewhere on the sphere of radius  $3/2$ , quadratic domination  $L_1 + L_2 \leq C_0\|w\|^2$  near 0, and a continuous compactly supported bump  $\psi \equiv 1$  there, the pencil difference obeys the  $t^{1-m-d/2}$  lower bound. The hypotheses are the multivariate analogues of the 1D turnkey theorem's, with analyticity in place of Taylor structure; homogeneity and continuity of  $P$  are not assumed but proven (multilinearity gives both).

## 3 Proof strategy

The deliberation (one consult round, immediate agreement on the target) proposed the classical route: expand  $g$  as a sum of diagonal terms, split off the first  $m + 1$ , and bound the tail by a geometric series using a coefficient bound at a radius inside the radius of convergence. While the consult ran, a direct search of Mathlib's analytic API found the uniform geometric approximation lemma<sup>4</sup>, which already packages the entire tail estimate: at radius  $r' < r$  it gives  $a \in (0, 1)$  and  $C > 0$  with  $\|f(x + y) - \sum_{k < n} p_k(y, \dots, y)\| \leq C(a\|y\|/r')^n$  for all  $n$ . Taking  $n = m + 1$ , the partial sum collapses to  $P(y)$  by the vanishing hypothesis (one `Finset.sum_eq_single_of_mem`),

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<sup>1</sup>2026-08-09-05-40-tide-germbij-turnkey-multi.

<sup>2</sup>`analytic_remainder_bound`.

<sup>3</sup>`analytic_pencil_difference_lower_bound_multi`.

<sup>4</sup>`HasFPowerSeriesOnBall.uniform_geometric_approx`.

and the remainder bound falls out with  $C' = C(a/r')^{m+1}$ ,  $u_1 = r'/4$ . The bridge is 18 lines. For the corollary, homogeneity of  $P$  is `ContinuousMultilinearMap.map_smul_univ` with the constant scaling vector (it holds for all  $c$ , so the turnkey's  $0 \leq c$  hypothesis is free), continuity is composition with the diagonal map, and the rest is application.

## 4 Roadblocks and resolutions

Two one-line fixes: the  $\mathbb{R}_{\geq 0}^\infty$  notation for the ball radius is scoped (needs `open scoped ENNReal NNReal`), and `positivity` cannot extract  $0 < a$  from an `Ioo` membership, so it must be surfaced as a hypothesis first. No thrashing, no consult beyond the deliberation round.

## 5 What was Mathlib, what was new

Almost all Mathlib: the uniform geometric approximation, the operator bound machinery behind it, `map_smul_univ`, and the partial-sum API. New: the observation that the approximation lemma at a *specific* truncation order, combined with a diagonal-vanishing hypothesis, is exactly a Taylor remainder bound; and the packaging that lets the `germbij` chain consume it.

## 6 Lessons

When a consult sketches a multi-step classical argument, spend five minutes checking whether Mathlib packages the composite before implementing the steps: the packaged lemma here absorbed the coefficient bound, the `HasSum` manipulation, and the geometric summation in one call. Searching for the *conclusion shape* (approximation by partial sums) found it; searching for the ingredients (coefficient bounds, tail sums) would have reproduced the manual route.

## 7 Follow-ups

- The hypothesis interface could be softened from `HasFPowerSeriesOnBall` to `AnalyticAt` (obtain the ball, then apply); a candidate for a future micro-tide only if a consumer wants it.
- Note annotation: tag `germbij` Theorem 7.3 with the composed corollary once merged.
- The diagonal-vanishing hypothesis is implied by (but weaker than) vanishing of the lower coefficients themselves; connecting it to `iteratedFDeriv` data would align with how a smooth-category consumer states assumptions.